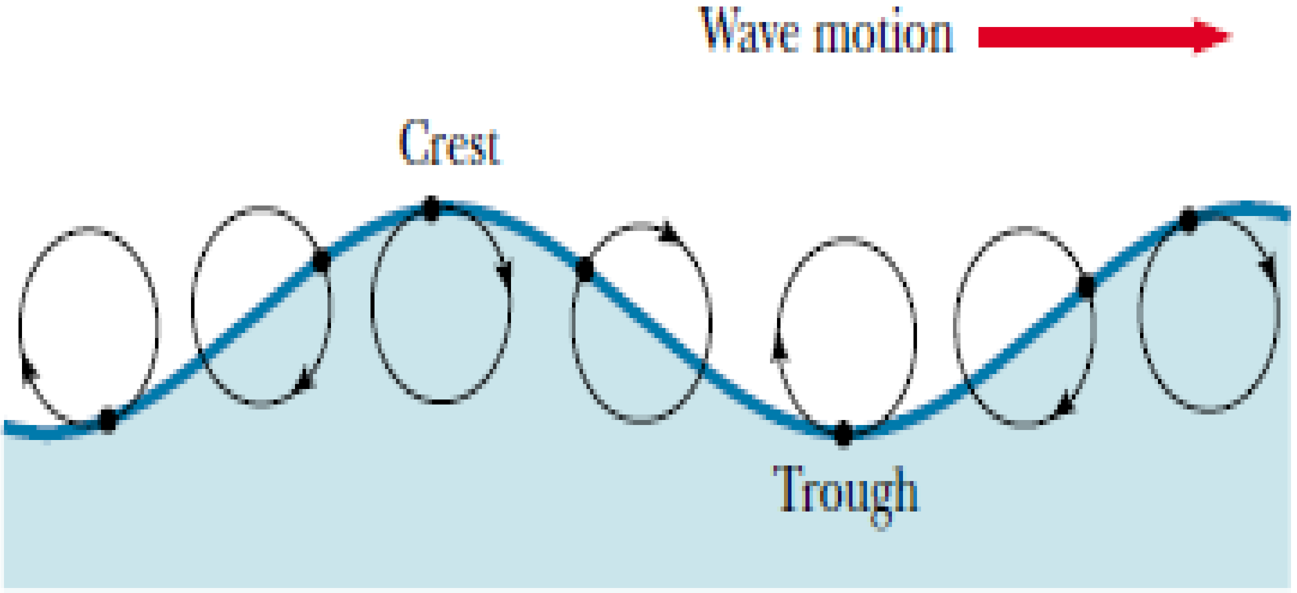
What is a wave ?

- Nature of waves:
  - ➔A wave is a traveling disturbance that transports energy from place to place.
  - ➔There are two basic types of waves: transverse and longitudinal.
  - ➔ Transverse: the disturbance travels perpendicular to the direction of travel of the wave.
  - ➔ Longitudinal: the disturbance occurs parallel to the line of travel of the wave.


Wave motion
Crest
Trough

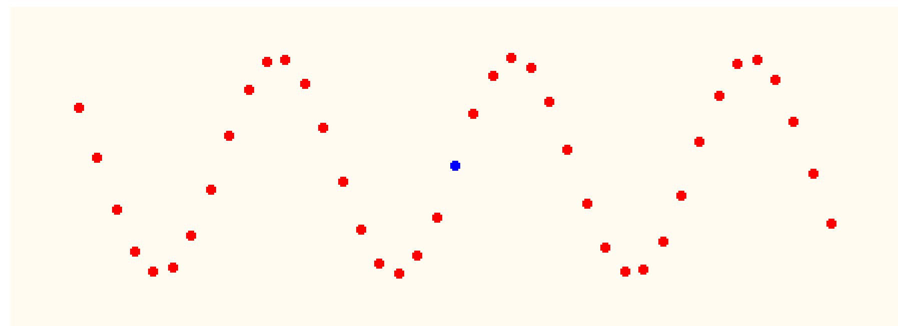
- Examples:
  - ➔ Longitudinal: Sound waves (e.g. air moves back & forth)
  - ➔ Transverse: Light waves (electromagnetic waves, i.e. electric and magnetic disturbances)

The source of the wave, i.e. the disturbance, moves continuously in

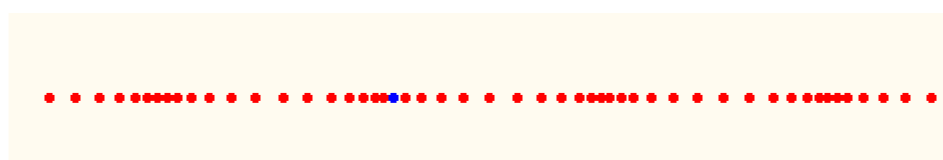
simple harmonic motion, generating an entire wave, where each part of the wave also performs a simple harmonic motion.

Types of Waves

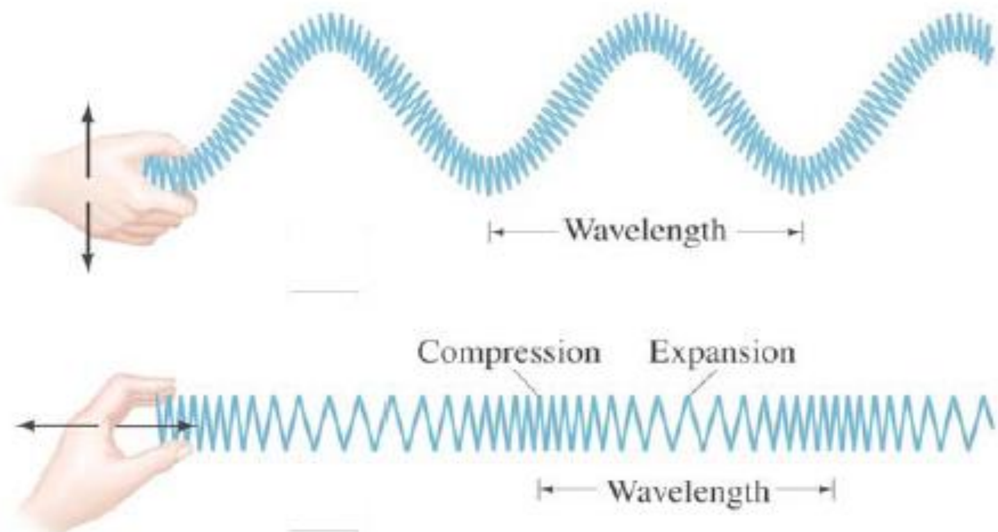
- **Transverse:** The medium oscillates perpendicular to the direction the wave is moving.
  - ➔ Water waves



- **Longitudinal:** The medium oscillates in the same direction as the wave is moving
  - ➔ Sound



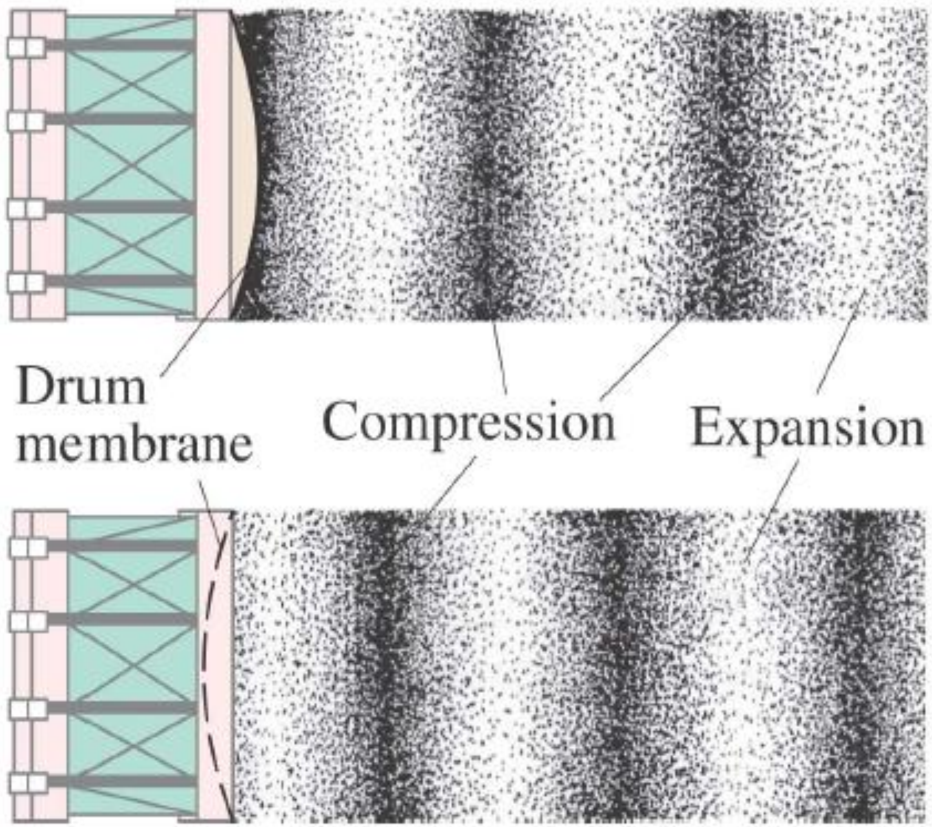
Types of Waves: Transverse and Longitudinal



The motion of particles in a wave can be either perpendicular to the wave direction (transverse) or parallel to it (longitudinal).

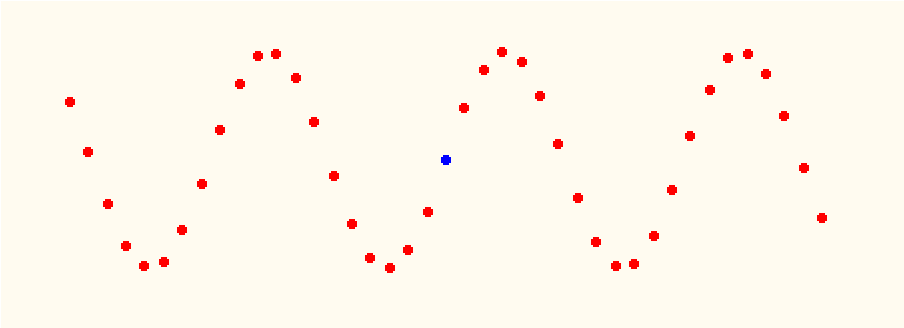
Types of Waves: Transverse and Longitudinal

Sound waves are longitudinal waves:



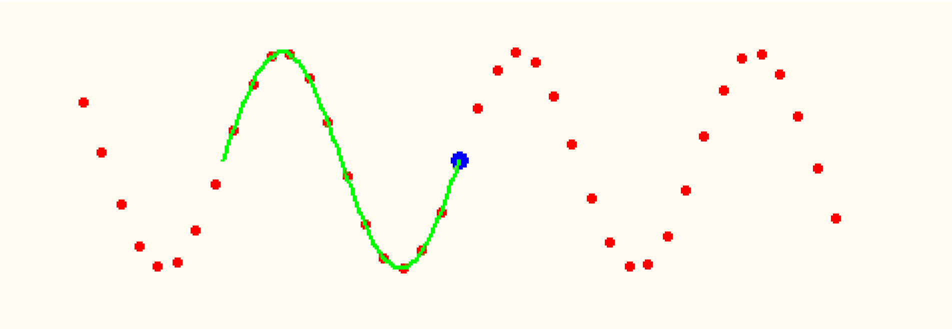
Wave Properties...

- Period: The time T for a point on the wave to undergo one complete oscillation.



- Speed: The wave moves one wavelength λ in one period T so its speed is $v = \lambda / T$.

$$\boxed{v = \frac{\lambda}{T}}$$



Wave Properties...

- The speed of a wave is a ***constant*** that depends only on the medium, not on the amplitude, wavelength or period:

  λ and T are related !

  $\lambda = vT$ or $\lambda = 2\pi v / \omega$ (since $T = 2\pi / \omega$)

  or $\lambda = v / f$ (since $T = 1/ f$)

- Recall f = cycles/sec or revolutions/sec

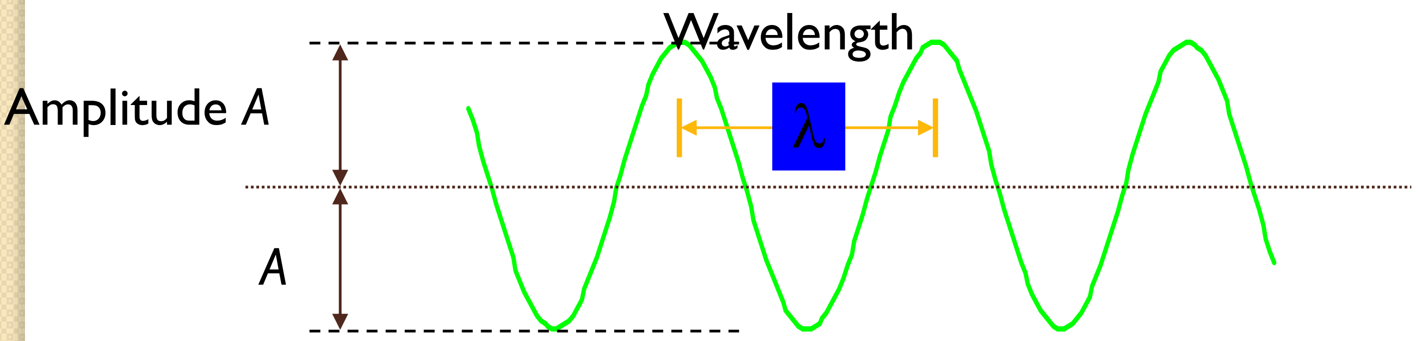
  $\omega = 2\pi f$

Is the speed of a wave particle the same as the speed of the wave ?

No. Wave particle performs simple harmonic motion: $v = A\,\omega \sin \omega t$.

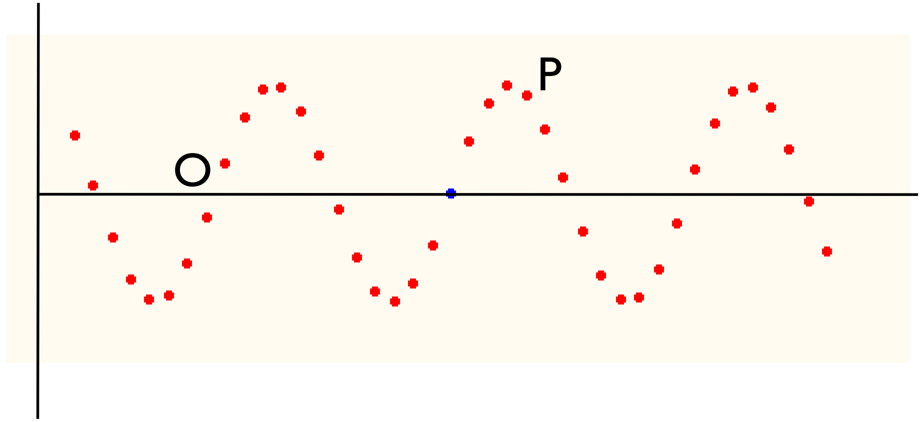
Wave Properties

- Amplitude: The maximum displacement *A* of a point on the
- wave.
- Wavelength: The distance λ between identical points
- on the wave.



Equation for a Progressive Wave

The simplest type of wave is the one in which the particles of the medium are set into simple harmonic vibrations as the wave passes through it. The wave is then called a simple harmonic wave.



Consider a particle O in the medium.
The displacement at any instant of time is given by

$$y = A \sin \omega t \ldots\ldots.(1)$$

Where A is the amplitude, ω is the angular frequency of the wave. Consider a particle P at a distance x from the particle O on its right. Let the wave travel with a velocity v from left to right. Since it takes some time for the disturbance to reach P, its displacement can be written as

$y = A \sin(\omega t - \phi)$.........(2)

Where ϕ is the phase difference between the particles O and P.

We know that a path difference of λ corresponds to a phase difference of 2π radians. Hence a path difference of x corresponds to a phase difference of

$$\frac{2\pi}{\lambda} . x$$

$$\phi = \frac{2\pi x}{\lambda}$$

Displacement of particle P is

$$y = A\sin(\omega t - \frac{2\pi x}{\lambda})..........(3)$$

$$\text{But } \omega = \frac{2\pi}{T}$$

$$y = A\sin(\frac{2\pi t}{T} - \frac{2\pi x}{\lambda})..........(5)$$

$$y = A\sin 2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right)$$

But $v = \frac{\lambda}{T}$ or $T = \frac{\lambda}{v}$

$$y = A\sin 2\pi\left(\frac{vt}{\lambda} - \frac{x}{\lambda}\right)$$

$$y = A\sin\frac{2\pi}{\lambda}(vt - x)\ldots\ldots\ldots\ldots(6)$$

Similarly, for a particle at a distance x to the left of 0, the equation for the displacement is given by

$$y = A\sin\frac{2\pi}{\lambda}(vt + x)\ldots\ldots\ldots\ldots\ldots(7)$$

Differential equation for wave motion

We have wave equation

$$y = A\sin\frac{2\pi}{\lambda}(vt - x)\ldots\ldots\ldots(1)$$

Differentiating equation with respect to x,
We get,

$$\frac{dy}{dx} = A\frac{2\pi}{\lambda}\cos\frac{2\pi}{\lambda}(vt - x)\ldots\ldots\ldots.(2)$$

$\frac{dy}{dx}$ represents the strain or the compression. When $\frac{dy}{dx}$ is positive, a rarefaction takes place and when $\frac{dy}{dx}$ is negative, a compression takes place.

The velocity of the particle whose displacement y is represented by equation, is obtained by differentiating it with respect to t, since velocity is the rate of change of displacement with respect to time.

$$\frac{dy}{dt} = A\frac{2\pi v}{\lambda}\cos\frac{2\pi}{\lambda}(vt - x)\text{.........}(3)$$

Comparing equations (2) and (3) we get,

$$\frac{dy}{dt} = v\frac{dy}{dx}\text{.........}(4)$$

Particle velocity = wave velocity x slope of the displacement curve or strain.

Differentiating equation (2)

$$\frac{d^2y}{dx^2} = -A\frac{4\pi^2}{\lambda^2}\sin\frac{2\pi}{\lambda}(vt - x)\text{..........}(5)$$

Differentiating equation (3)

$$\frac{d^2 y}{dt^2} = -A\frac{4\pi^2 v^2}{\lambda^2}\sin\frac{2\pi}{\lambda}(vt - x)\text{.........}(6)$$

Comparing equs. (5) and (6)

$$\frac{d^2 y}{dt^2} = v^2\frac{d^2 y}{dx^2}\text{.........}(7)$$

Equation (7) represents the differential equation of wave velo0city

Ex. The equation of a traveling wave is

$$y = 4.0 \sin \pi (0.10x - 2t)$$

Find (i) wavelength, (ii) speed and (iii) frequency of oscillating particle of the wave

Ex. When a simple harmonic wave is propagated through a medium, the displacement of a paricle in cm at any instant is

$$y = 10\sin\frac{2\pi}{100}(36000t - 20)$$

Calculate the amplitude, wave velocity, wavelength, frequency and period of the oscillating particle.

- Ex. The equation of a traveling wave is

$$y = 4.0 \sin \pi (0.10x - 2t)$$

Find (i) amplitude (ii) wavelength (iii) speed (iv) frequency of wave

When a simple harmonic wave is propagated through a medium, the displacement of the particle at any instant of time is given by

$$y = 5.0 \sin \pi (360t - 0.15x)$$

calculate

(i)the amplitude of the vibrating particle,

(ii)wave velocity,

(ii)wave length,

(iv)frequency and

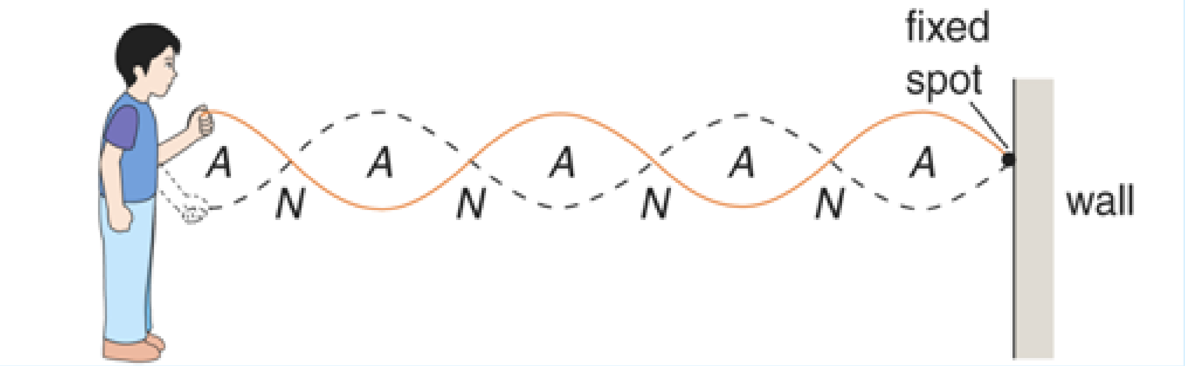
(v) time period.

Ex. A simple harmonic wave of amplitude 8units travels a line of particles in the direction of positive X axis. At any instant for a particle at a distance of 10cm from the origin, the displacement is +6units and at a distance a particle from the origin is 25units, the displacement is +4units. Calculate the wavelength.

Introduction to stationary waves

Stationary waves (standing waves) produced in
- string of guitar or piano
- rope (one end tie to wall)

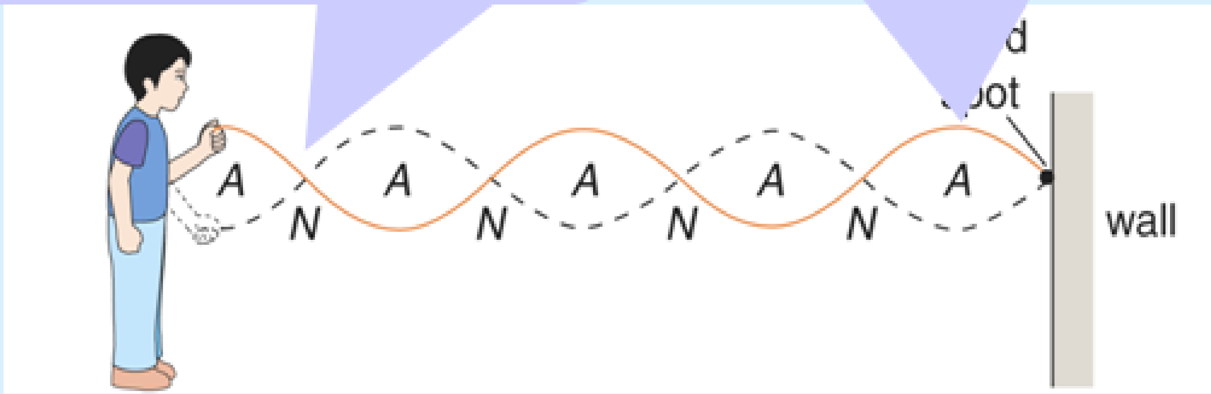
- Incident wave is reflected by the wall to from reflected wave
- superposition of the waves produces stationary wave



Introduction to stationary waves

N – nodes (remain stationary)

A – antinodes (largest amplitude of oscillation)



A stationary wave is produced by the superposition of two progressive waves of the same amplitude and frequency but travelling in opposite directions.

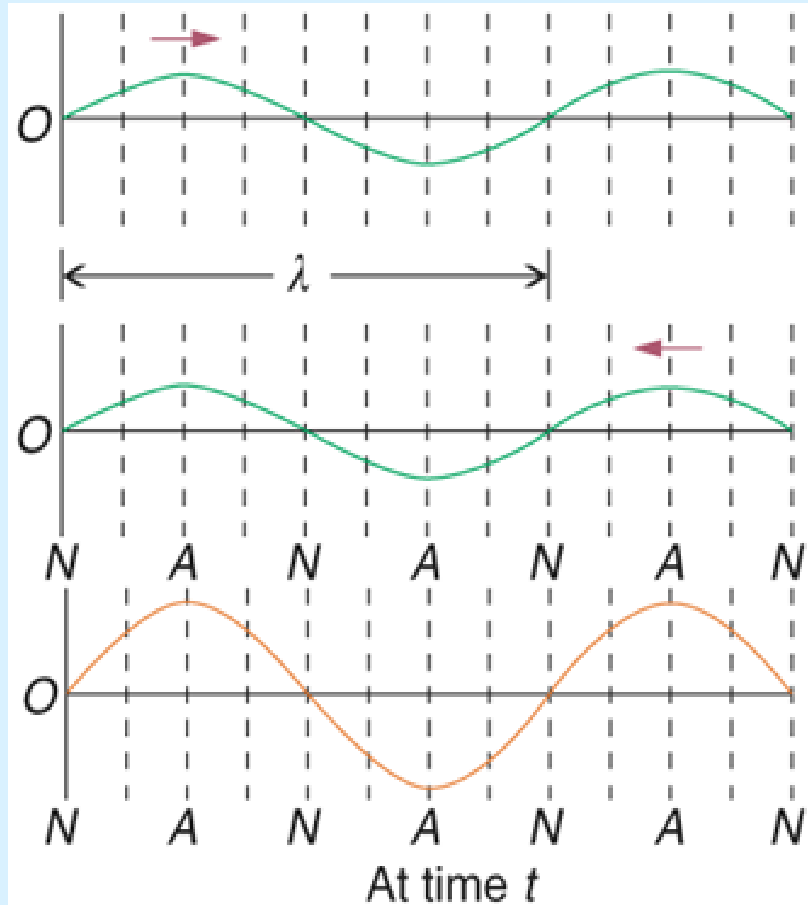
Introduction to stationary waves

(a) at time t (constructive interference)

(1) To right

(2) To left

(1) + (2) = (3)



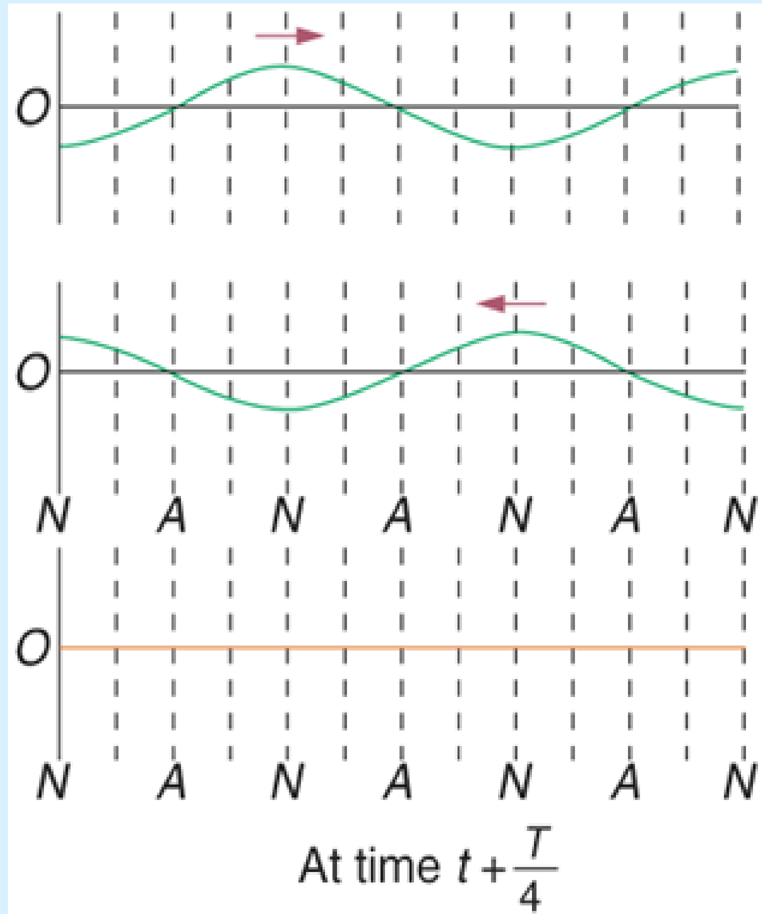
Introduction to stationary waves

(b) at time $t + T/4$ (destructive interference)

(1)

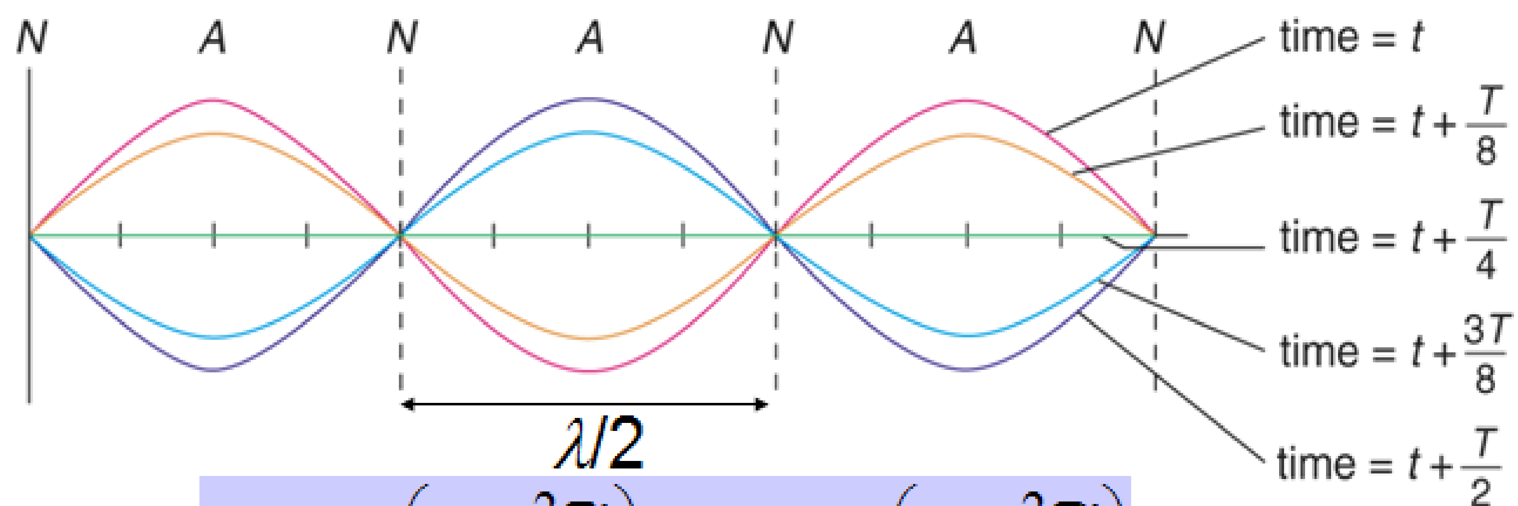
(2)

(1) + (2) = (3)



Introduction to stationary waves

Vibration of stationary wave at various instants



$$y_1 = a\sin\left(\omega t - \frac{2\pi x}{\lambda}\right), \quad y_2 = a\sin\left(\omega t + \frac{2\pi x}{\lambda}\right)$$

$$y = y_1 + y_2 = \left(2a\cos\frac{2\pi x}{\lambda}\right)\sin\omega t$$

Amplitude (A) of resultant stationary wave

The positions of antinodes

Progressive wave	Stationary wave
Energy is transferred along the direction of propagation.	No energy is transferred along the direction of propagation.
The wave profile moves in the direction of propagation.	The wave profile does not move in the direction of propagation.
Every point along the direction of propagation is displaced.	There are points known as nodes where no displacement occurs.
Every point has the same amplitude.	Points between two successive nodes have different amplitudes.
Neighbouring points are not in phase.	All points between two successive nodes vibrate in phase with one other.

Forced Oscillations; Resonance

Forced vibrations occur when there is a periodic driving force. This force may or may not have the same period as the natural frequency of the system.

If the frequency is the same as the natural frequency, the amplitude can become quite large. This is called resonance.

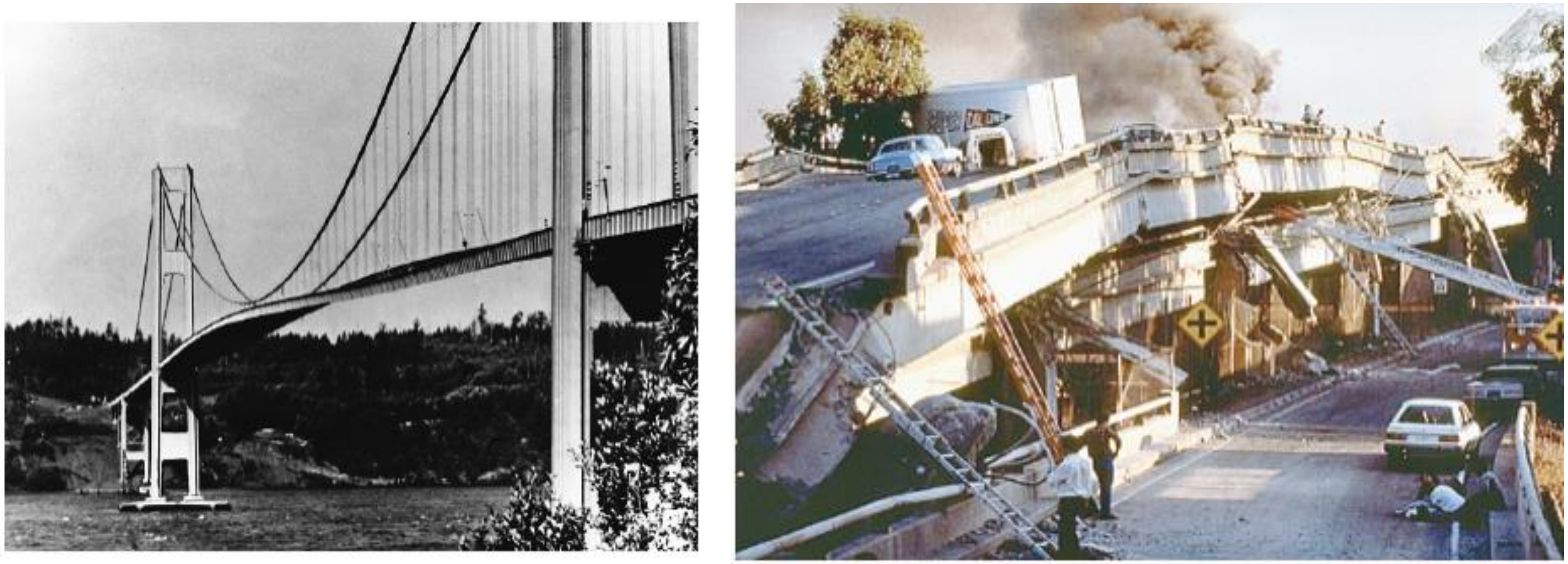
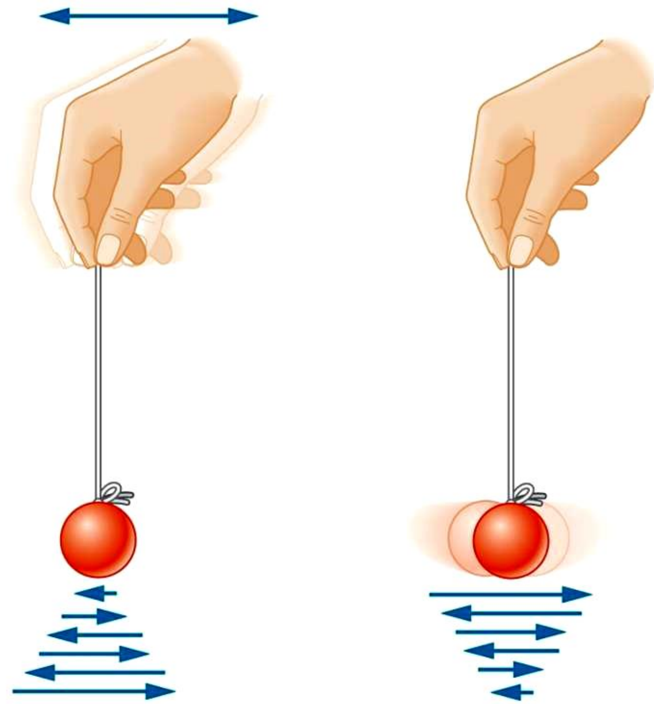


Figure : Caption: (a) Large-amplitude oscillations of the Tacoma Narrows Bridge, due to gusty winds, led to its collapse (1940). (b) Collapse of a freeway in California, due to the 1989 earthquake.


m



Equation of Driven Harmonic Motion.

- Consider what happens when we apply a time-dependent force $F(t)$ to a system that normally would carry out SHM with an angular frequency ω_0.

- Assume the external force $F(\mathrm{t}) = mF_0\sin(\omega t)$. The equation of motion can now be written as

$$\frac{d^2x}{dt^2} = -\omega_0{}^2 x + F_0 \sin\left(\omega t\right)$$

- The steady state motion of this system will be harmonic motion with an angular frequency equal to the angular frequency of the driving force.

- Consider the general solution

$$x(t) = A\cos(\omega t + \phi)$$

- The parameters in this solution must be chosen such that the equation of motion is satisfied. This requires that

$$-\omega^2 A\cos(\omega t + \phi) + \omega_0{}^2 A\cos(\omega t + \phi) - F_0 \sin(\omega t) = 0$$

- This equation can be rewritten as

$$\left(\omega_0{}^2 - \omega^2\right) A\cos(\omega t)\cos(\phi) -$$
$$\left(\omega_0{}^2 - \omega^2\right) A\sin(\omega t)\sin(\phi) - F_0 \sin(\omega t) = 0$$

- Our general solution must thus satisfy the following condition:

$$\left(\omega_0^2-\omega^2\right)A\cos\left(\omega t\right)\cos\left(\phi\right)-\left\{\left(\omega_0^2-\omega^2\right)A\sin\left(\phi\right)-F_0\right\}\sin\left(\omega t\right)=0$$

- Since this equation must be satisfied at all time, we must require that the coefficients of cos(ωt) and sin(ωt) are 0. This requires that

$$\left(\omega_0^2-\omega^2\right)A\cos\left(\phi\right)=0$$

and

$$\left(\omega_0^2-\omega^2\right)A\sin\left(\phi\right)-F_0=0$$

- The interesting solutions are solutions where $A \neq 0$ and $\omega \neq \omega_0$. In this case, our general solution can only satisfy the equation of motion if

$$\cos(\phi) = 0$$

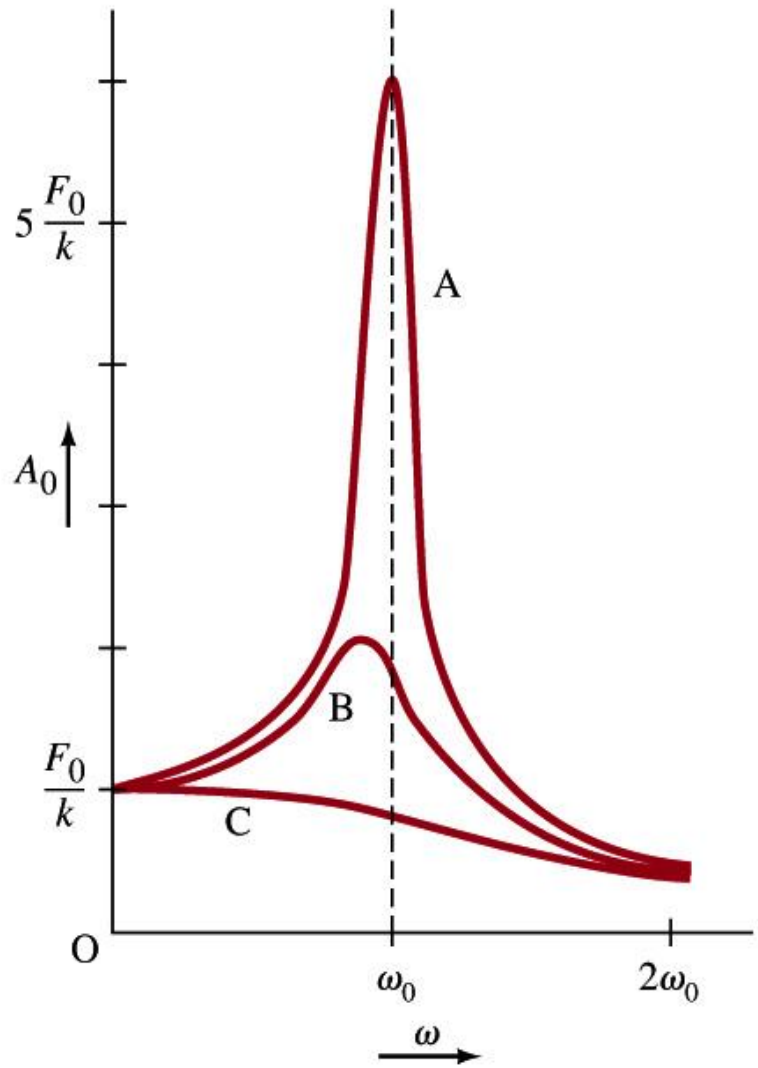
and

$$(\omega_0{}^2 - \omega^2)A\sin(\phi) - F_0 = (\omega_0{}^2 - \omega^2)A - F_0 = 0$$

- The amplitude of the motion is thus equal to

$$A = \frac{F_0}{(\omega_0{}^2 - \omega^2)}$$

- If the driving force has a frequency close to the natural frequency of the system, the resulting amplitudes can be very large even for small driving amplitudes. The system is said to be in resonance.
- In realistic systems, there will also be a damping force. Whether or not resonance behavior will be observed will depend on the strength of the damping term.



$5\,\frac{F_0}{k}$
A_0
$\frac{F_0}{k}$
A
B
C
O
ω_0
$2\omega_0$
ω

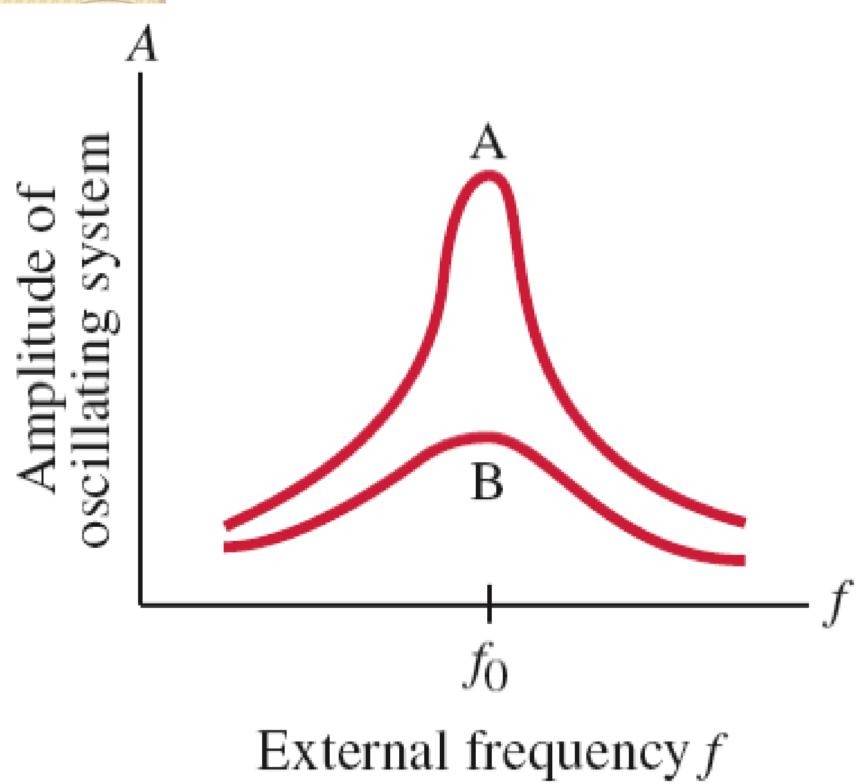
Driven Harmonic Motion.

Done for today!
Thursday: Temperature and Heat!

QuickTime™ and a
TIFF (Uncompressed) decompressor
are needed to see this picture.

Unusually Strong Cyclone Off the Brazilian Coast: A lot of Rotational Motion!
Credit: Jacques Descloitres, MODIS Land Rapid Response Team, GSFC, NASA

Forced Oscillations; Resonance



The sharpness of the resonant peak depends on the damping. If the damping is small (A) it can be quite sharp; if the damping is larger (B) it is less sharp.

Like damping, resonance can be wanted or unwanted. Musical instruments and TV/radio receivers depend on it.

Forced Oscillations; Resonance

The equation of motion for a forced oscillator is:

$$\boxed{ma = -kx - bv + F_0 \cos \omega t.}$$

The solution is: $$x = A_0 \sin(\omega t + \phi_0),$$

where $$A_0 = \frac{F_0}{m\sqrt{(\omega^2 - \omega_0^2)^2 + b^2\omega^2/m^2}}$$

and

$$\phi_0 = \tan^{-1} \frac{\omega_0^2 - \omega^2}{\omega(b/m)}.$$

Forced Oscillations; Resonance

The width of the resonant peak can be characterized by the Q factor:

$$Q = \frac{m\omega_0}{b}.$$